Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 5:** Absolute Value Equations

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $|x| = 3$

$x = 3$ $\quad x = -3$

$x = \{-3, 3\}$

2. $|m| = -10$

$\emptyset$

3. $|-6k| = 30$

$\frac{-6k}{-6} = \frac{30}{-6}$ $\quad \frac{-6k}{-6} = \frac{-30}{-6}$

$k = -5$ $\quad k = 5$

$k = \{-5, 5\}$

4. $|c - 10| = 3$

$c - 10 = 3$ $\quad c - 10 = -3$

$+10 \ +10$ $\quad +10 \ +10$

$c = 13$ $\quad c = 7$

$c = \{7, 13\}$

5. $|2k + 3| = 7$

$2k + 3 = 7$ $\quad 2k + 3 = -7$

$-3 \ -3$ $\quad -3 \ -3$

$\frac{2k}{2} = \frac{4}{2}$ $\quad \frac{2k}{2} = \frac{-10}{2}$

$k = 2$ $\quad k = -5$

$k = \{-5, 2\}$

6. $|-4g - 2| = 10$

$-4g - 2 = 10$ $\quad -4g - 2 = -10$

$+2 \ +2$ $\quad +2 \ +2$

$\frac{-4g}{-4} = \frac{12}{-4}$ $\quad \frac{-4g}{-4} = \frac{-8}{-4}$

$g = -3$ $\quad g = 2$

$g = \{-3, 2\}$

7. $\left|\frac{1}{2}x + 5\right| = 13$

$\frac{1}{2}x + 5 = 13$ $\quad \frac{1}{2}x + 5 = -13$

$-5 \ -5$ $\quad -5 \ -5$

$2 \cdot \frac{1}{2}x = 8 \cdot 2$ $\quad 2 \cdot \frac{1}{2}x = -18 \cdot 2$

$x = 16$ $\quad x = -36$

$x = \{-36, 16\}$

8. $|26 - 4c| = -2$

$\emptyset$

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